

Learning Objective 1.4: I can calculate input impedance, feed considerations, and the role of baluns in an antenna feed system.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Radiation resistance and efficiency.** A short dipole has length $\ell = 0.08\lambda$. Its conductors contribute a loss resistance of $R_{\text{loss}} = 1.5\Omega$.

(a) Compute the radiation resistance $R_{\text{rad}} = 80\pi^2(\ell/\lambda)^2$.

$$R_{\text{rad}} = 80\pi^2(0.08)^2 = 789.6 \times 0.0064 \approx 5.05\Omega$$

$$R_{\text{rad}} = \approx 5.05\Omega$$

(b) Compute the radiation efficiency η_{rad} .

$$\eta_{\text{rad}} = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}} = \frac{5.05}{6.55} \approx 0.77$$

$$\eta_{\text{rad}} = \approx 0.77 \text{ (77\%)}$$

(c) The short dipole's directivity is $D = 1.5$ (1.76 dBi). Find the gain in dBi.

$$G_{\text{dBi}} = 1.76 + 10\log_{10}(0.77) = 1.76 - 1.13 \approx 0.6 \text{ dBi}$$

$$G = \approx 0.6 \text{ dBi}$$

2. Feeding a half-wave dipole with 50Ω coax.

- (a) A dipole trimmed to resonance presents $Z_{\text{in}} = 70 + j0\Omega$. Find Γ , VSWR, return loss, and mismatch loss on a 50Ω line.

$$\Gamma = \frac{70 - 50}{70 + 50} = 0.167, \quad \text{VSWR} = \frac{1.167}{0.833} \approx 1.40$$

$$\text{RL} = -20\log_{10}(0.167) \approx 15.6 \text{ dB}, \quad L_{\text{mm}} = -10\log_{10}(1 - 0.167^2) \approx 0.12 \text{ dB}$$

$$\text{VSWR} = \approx 1.40$$

$$\text{RL} = \approx 15.6 \text{ dB}$$

- (b) An *untrimmed* half-wave dipole presents $Z_{\text{in}} = 73 + j42.5\Omega$. Find $|\Gamma|$ and VSWR on the same line.

$$|\Gamma| = \frac{\sqrt{23^2 + 42.5^2}}{\sqrt{123^2 + 42.5^2}} = \frac{48.3}{130.1} \approx 0.371$$

$$\text{VSWR} = \frac{1.371}{0.629} \approx 2.18$$

$$|\Gamma| = \approx 0.371$$

$$\text{VSWR} = \approx 2.18$$

- (c) Compare (a) and (b): in one sentence, what did trimming to resonance buy you, and why?

Trimming killed the $+j42.5\Omega$ reactance, dropping VSWR from ≈ 2.2 to 1.4 — the reactance, not the resistance, was doing most of the damage.

3. **Quarter-wave transformer.** A resonant antenna presents a real input impedance $R_L = 36\Omega$. You want to match it to a 50Ω feed line with a quarter-wave transformer.

- (a) What characteristic impedance Z_1 must the $\lambda/4$ section have?

$$Z_1 = \sqrt{Z_0 R_L} = \sqrt{(50)(36)} = \sqrt{1800} \approx 42.4\Omega$$

$$Z_1 = \boxed{\approx 42.4\Omega}$$

- (b) What is Γ at the design frequency?

The transformer makes $Z_1^2/R_L = 1800/36 = 50\Omega = Z_0$, so $\Gamma = 0$.

- (c) In one sentence, what happens to the match off the design frequency, and why?

The section is $\lambda/4$ only at the design frequency; off frequency its electrical length is no longer 90° , the transformation Z_1^2/R_L no longer holds, and $|\Gamma|$ climbs — the match is narrowband.

4. **VSWR, reflection, and the impedance behind it.** An antenna measures $\text{VSWR} = 1.5$ on a 50Ω line.

(a) Find $|\Gamma|$.

$$|\Gamma| = \frac{1.5 - 1}{1.5 + 1} = \frac{0.5}{2.5} = 0.20$$

$$|\Gamma| = 0.20$$

(b) Find the return loss in dB.

$$\text{RL} = -20\log_{10}(0.20) \approx 14 \text{ dB}$$

$$\text{RL} = \approx 14 \text{ dB}$$

(c) Find the mismatch loss in dB.

$$L_{\text{mm}} = -10\log_{10}(1 - 0.20^2) = -10\log_{10}(0.96) \approx 0.18 \text{ dB}$$

$$L_{\text{mm}} = \approx 0.18 \text{ dB}$$

(d) Assuming a purely resistive antenna, what two real Z_{in} are consistent with this VSWR?

$$Z_{\text{in}} = 50 \cdot \frac{1.2}{0.8} = 75\Omega \quad \text{or} \quad 50 \cdot \frac{0.8}{1.2} = 33.3\Omega$$

$$Z_{\text{in}} = 75 \text{ or } 33.3\Omega$$

VSWR alone does not say which side of Z_0 you are on.

5. **Mismatch that kills the amplifier.** A power amplifier (PA) delivers a forward power of $P_{\text{fwd}} = +40$ dBm toward a transmit antenna. A shunt protection diode at the PA output fails catastrophically if it must dissipate more than $P_{\text{max}} = +30$ dBm. Model *all* of the reflected power as landing on the diode.

- (a) Convert P_{fwd} and P_{max} to watts.

$$P_{\text{fwd}} = 10^{(40-30)/10} \text{W} = 10 \text{W}, \quad P_{\text{max}} = 10^0 \text{W} = 1 \text{W}$$

- (b) In terms of $|\Gamma|$ at the antenna, how much power is reflected back toward the PA?

$$P_{\text{ref}} = |\Gamma|^2 P_{\text{fwd}}.$$

- (c) Find the largest $|\Gamma|$ the system can tolerate before the diode is destroyed.

$$|\Gamma|_{\text{max}} = \sqrt{\frac{P_{\text{max}}}{P_{\text{fwd}}}} = \sqrt{0.1} \approx 0.316$$

- (d) Convert to a VSWR limit and a return-loss limit, and state the datasheet rule.

$$\text{VSWR}_{\text{max}} = \frac{1.316}{0.684} \approx 1.93, \quad \text{RL}_{\text{min}} = -10 \log_{10}(0.1) = 10 \text{ dB}$$

Rule: keep $\text{VSWR} \leq 1.9$ (return loss ≥ 10 dB); reflected power must stay ≥ 10 dB below forward.

- (e) In one sentence: how does a circulator or isolator between the PA and antenna change this picture?

It routes the reflected wave into a matched dump-port load instead of back into the PA, so the diode no longer sees it — the antenna can then run at much higher VSWR (paid for by insertion loss, size, and bandwidth).

6. Baluns.

- (a) In one or two sentences, explain why feeding a dipole *directly* with coax (no balun) distorts the pattern.

Coax is unbalanced, so the two arms present unequal impedances to it. The imbalance drives a common-mode current on the *outside* of the shield; that shield current radiates, so the feed line becomes part of the antenna and skews the pattern.

- (b) You must feed a folded dipole ($Z_{\text{in}} \approx 292\Omega$) with 75Ω coax. Which balun would you choose, and what does it do?

A 4:1 (half-wave) balun: it balances the feed *and* transforms impedance 4:1, taking 292Ω down to $\approx 73\Omega$ — a good match to 75Ω coax.

- (c) You feed a straight resonant dipole ($\approx 70\Omega$) with 50Ω coax. Which balun would you choose, and why is it a *different* choice than (b)?

A 1:1 current (choke) balun. The 70Ω dipole already matches 50Ω well ($\text{VSWR} \approx 1.4$), so no impedance transformation is wanted — you only need to choke off the shield current. A 4:1 balun here would ruin the match.

Documentation: _____